Fiscal Year:	FY 2009	**Task Last Updated:**	FY 06/22/2010
PI Name:	McCurdy, Michael M.S.		
Project Title:	Crew Scheduling Tools		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ludowise, Melissa (NASA Ames Research Center) Marquez, Jessica (NASA Ames Research Center) Li, Jack (San Jose State University Foundation)		
Grant/Contract No.:			
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Task Description:	Note Task added to Task Book in April 2009. Future mission concepts will require a significantly more efficient planning process and tools. The Science Planning Interface (SPIFe) toolkit and underlying planning and scheduling technology has made significant advancements toward this goal. Given past successes in the Mars program, Space Human Factors Engineering has identified this as an ideal project to showcase the value of a data-driven user-centered design process that will yield tangible results in the near term and accomplish the needed capability of on-board re-planning in the future. In recent years, Mars surface missions operated by NASA's Jet Propulsion Laboratory (JPL) have made significant advancements in activity planning tools and processes. By many accounts, the efficiency with which plans are generated for these missions has improved by an order of magnitude over the previous baseline. The Science Planning Interface (SPIFe) toolkit developed at Ames Research Center and the human-centered design process employed by the Ames Human-Computer Interaction Group has been central to this improvement. Many of the efficiency gains discovered and implemented for the Mars program were the direct result of human factors analysis and tool design and not improvements to the underlying planning technology. The activity planning process used today on Shuttle and ISS is similar to the process used by JPL for deep space robotic mission planning – there is limited automated planning support and constraint and resource checking capability. As execution time approaches, it is increasingly costly to make needed changes to schedules. There is little to no real-time re-planning capability either on the ground or on-board. In FY07, we undertook an effort to adapt several core SPIFe components for use in planning and scheduling operations for the Flight Analogs Project Bedrest study. The collaboration was successful on both sides: the Bedrest study provided a real context in which to test and deploy the SPIFe next-generation planning tools, and we created a partially automated planning systems for Flight Analogs scheduling personnel use that is measurably better than the baseline Excel-based approach in a number of ways. Bedrest study scheduling personnel have begun migrating to SPIFe as their primary tool for tactical scheduling. In FY08, we continued the collaboration with the Bedrest team, focusing on proposed enhancements to the core functionality that will enhance the utility of SPIFe across the board. The majority of the research and design work for the Bedrest domain was completed. In FY09, we will begin another adaptation for a domain relevant to the Constellation Program. Through extensive user research in this new domain, we will further broaden our understanding of how users interact with planning systems in operational environments. Additionally, we will focus on documenting generalizable lessons learned to date across the Mars, Bedrest, and NEEMO deployments of SPIFe.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although the focus of this research was planning for space missions, many of the same principles apply to ground-based planning and execution processes. Lessons learned from analysis of four planning domains and the resulting tool solutions could be applied to benefit life on Earth, though we have not pursued these applications.
Task Progress:	Beginning with our experience developing tools for the Mars Exploration Rover mission in 2003, the authors of this paper have been actively involved in researching and advancing the state-of-the art in planning and scheduling tools for NASA mission operations. We have added three additional planning and scheduling domains to our initial Mars experience: The Flight Analogs Project Bedrest Study, NASA Extreme Environment Mission Operations, and Operations Planning for the International Space Station. In this paper, we outline some of the key similarities and differences that we have observed between these domains. We then discuss some of the key insights and lessons learned from observing and participating in the activity planning process in these domains, and we make some attempts to generalize these findings more broadly. We have also been actively involved in designing and guiding the development of a planning toolkit, the Scheduling and Planning Interface for Exploration, that has served both as a rich prototyping environment in which to test our design solutions as well as an operational tool supporting multiple flight projects across the agency. In this paper, we outline several unique solutions that we have implemented within the toolkit, and we tie them to our research findings. Although we have had an opportunity to learn second-hand from several planning domains in industry and academia, this report will focus on the domains within NASA with which we have been actively involved. In each case, we have actively attempted to adapt our core planning software to support planners, with varying degrees of success. There are four such domains, which we will discuss further in the following sections — science activity planning for current and future Mars surface missions, medical operations planning for a long-duration microgravity experiment facility operated by the Flight Analogs Project Bed Rest Study out of Johnson Space Center (JSC), operations planning for the NASA Extreme Environment Mission Operations (NEEMO) analog, and finally operations planning for crew and ground operations onboard the International Space Station (ISS). We have characterized the activity of planning for these domains thoroughly in terms of key similarities, differences, and potential areas where tools can be improved to better meet mission needs. We have taken key lessons learned and modified our suite of planning tools accordingly to better support planning work in practice, and to assess the effectiveness of various software technology and process solutions.
Bibliography Type:	Description: (Last Updated:)